

UNIT II

LINEAR DATA STRUCTURES

PART A

Q.NO	Question	Bloom's Taxonomy Level															
1	Point out the advantage of representing stack using a linked list than array. ● It is not necessary to specify the number of elements to be stored in a stack during its declaration, since memory is allocated dynamically at run time when an element is added to the stack. ● Insertions and deletions can be handled easily and efficiently. ● Linked list representation of stacks can grow and shrink in size without wasting memory space, depending upon the insertion and deletion that occurs in the list. ● Multiple stacks can be represented efficiently using a chain for each stack.	K1															
2	Point out the rules followed during the infix to postfix conversions. ● Fully parenthesize the expression starting from left to right. During parenthesizing, the operators having higher precedence are first parenthesized. ● Move the operators one by one to their right, such that each operator replaces their corresponding right parenthesis. ● The part of the expression, which has been converted into postfix is to be treated as single operand.	K1															
3	Compare the working of stack and queue data structure. <table><tr><th>Basis for comparison</th><th>Stack</th><th>Queue</th></tr><tr><td>Working principle</td><td>LIFO (Last in First out)</td><td>FIFO (First in First out)</td></tr><tr><td>Operations performed</td><td>Push and pop</td><td>Enqueue and dequeue</td></tr><tr><td>Examination of initial condition</td><td>Top == -1</td><td>Front == -1 front = rear +1</td></tr><tr><td>Examination of full condition</td><td>Top == max -1</td><td>Rear == max -1</td></tr></table>	Basis for comparison	Stack	Queue	Working principle	LIFO (Last in First out)	FIFO (First in First out)	Operations performed	Push and pop	Enqueue and dequeue	Examination of initial condition	Top == -1	Front == -1 front = rear +1	Examination of full condition	Top == max -1	Rear == max -1	K2
Basis for comparison	Stack	Queue															
Working principle	LIFO (Last in First out)	FIFO (First in First out)															
Operations performed	Push and pop	Enqueue and dequeue															
Examination of initial condition	Top == -1	Front == -1 front = rear +1															
Examination of full condition	Top == max -1	Rear == max -1															
4	Define stack.	K1															

	Stack is an ordered collection of elements in which insertions and deletions are restricted to one end. The end from which elements are added and/or removed is referred to as top of the stack. Stacks are also referred as piles, push-down lists and last-in-first-out (LIFO) lists.	
5	List any four applications of stack. • Towers of Hanoi • Reversing a string • Balanced parenthesis • Recursion using stack • Evaluation of arithmetic expressions	K1
6	Convert the Following Infix expression to postfix expression using stack $(a+b)*(c-d)/(e+f)$ Postfix Evaluation • $(a+b) * (c-d) / (e+f)$ • $a b + c d - * e f + /$ • $a b + c d - * e f + /$ • $a b + c d - * e f + /$ • $a b + c d - * e f + /$ • $a b + c d - * e f + /$ • $a b + c d - * e f + /$ d c $(a+b)$ stack	K3
6	State the different ways of representing expressions The different ways of representing expressions are • Infix Notation • Prefix Notation • Postfix Notation	K1
7	List the applications of a queue. • Jobs submitted to printer • Real life line • Calls to large companies • Access to limited resources in Universities • Accessing files from file server	K1
9	What are the advantages of using Linear Structures. The advantages of using Linear Structures are, • Efficient memory usage • Easy implementation • Fast search and access times.	K1
10	How does a stack differ from a queue in terms of linear structure? • Stack follows LIFO (Last In First Out), whereas queue follows FIFO (First In First Out) order. • Stack adds and removes element from the top, whereas queue adds elements to the end and removes from the front. • Stack is like a vertical pile of plates, whereas queue is like a horizontal line of people waiting.	K1

11	What is Linear Structure? A Linear structure is a data structure in which elements are arranged in a sequential manner, where each element has a unique predecessor and successor.	K1
12	State the difference between queues and linked lists The difference between queues and linked lists is that insertions and deletions may occur anywhere in the linked list, but in queues insertions can be made only in the rear end and deletions can be made only in the front end.	K2
13	Define Queue. Queue is an ordered collection of elements in which insertions are restricted to one end called the rear end and deletions are restricted to other end called the front end. Queues are also referred as First-In-First-Out (FIFO) Lists.	K1
14	State the difference between stacks and linked lists The difference between stacks and linked lists is that insertions and deletions may occur anywhere in a linked list, but only at the top of the stack	K2
15	What is data structure? Data Structures is defined as the way of organizing all data items that consider not only the elements stored but also stores the relationship between the elements.	K1
16	State the different types of linked lists. The different types of linked list include singly linked list, doubly linked list and circular linked list.	K1
17	Define an Abstract Data Type (ADT). An abstract data type is a set of operations. ADTs are mathematical abstractions; nowhere in an ADT's definition is there any mention of how the set of operations is implemented. Objects such as lists, sets and graphs, along with their operations can be viewed as abstract data types.	K1
18	What are the objectives of studying data structures? • To identify and create useful mathematical entities and operations to determine what classes of problems can be solved using these entities and operations • To determine the representation of these abstract entities and to implement the abstract operations on these concrete representation	K1
19	What are the various Operations performed on the Stack? The various operations that are performed on the stack are; • CREATE(S) – Creates S as an empty stack. • PUSH(S, X) – Adds the element X to the top of the stack. • POP(S) – Deletes the top most elements from the stack. • TOP(S) – returns the value of top element from the stack. • ISEMTPTY(S) – returns true if Stack is empty else false. • ISFULL(S) - returns true if Stack is full else false.	K1

20	How do you test for an empty Queue? The condition for testing an empty queue is rear=front-1. In linked list implementation of queue the condition for an empty queue is the header node link field is NULL.	K1
----	--	----

PART B

Q.NO	Question	Bloom's Taxonomy Level
1	Explain array based implementation of stacks. (8) Explain linked list implementation of stacks. (8)	K2
2	Describe about stack ADT in detail any one application of stack. (16)	K2
3	Write an algorithm to convert an infix expression to a postfix expression, and convert the infix expression $a + b * c - d / e ^ f$ to a postfix expression. (16)	K2
4	i) Give an algorithm for stack operations on using a linked list (8) ii) Discuss about addition and deletion operations performed on a circular queue with necessary algorithms. (8)	K2
5	Briefly describe the operations of queue with examples. (16)	K2
6	Write an algorithm to implement queue functions using arrays. (16)	K2
7	Evaluate the infix expression using stack and also convert the following expression to postfix: $3 + (4 * 2) - (10 / 2 - 3) ^ 2$. (16)	K2
8	Explain an algorithm to implement the circular queue using arrays. List the applications of Queues. (16)	K2
9	Show the simulation using stack for the following expression to convert infix to postfix : $(P+Q)*(R-S)/(T-U)+V*(W+X)$ (16)	K1